

Notes: Gabaix & Koijen (JPE, 2024)

Granular Instrumental Variables

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1 Big picture

- **Question.** How can economists construct valid instruments when aggregate outcomes are affected by idiosyncratic shocks to a few large firms, industries, countries, or investor sectors?
- **Main idea.** In granular economies, shocks to large units move aggregate variables. If those shocks are idiosyncratic and can be separated from common factors, they become valid instruments for endogenous aggregate regressors such as prices, quantities, exchange rates, or flows.
- **Contribution.** The paper proposes **granular instrumental variables (GIVs)**: weighted sums of idiosyncratic shocks, where the weights emphasize large units and purge common shocks.
- **Core application in the paper.** A supply–demand system in which idiosyncratic demand shocks to large countries or firms identify both the supply elasticity and the demand elasticity.
- **Main methodological punchline.** The optimal GIV is the projection of the size vector onto the subspace orthogonal to common-factor loadings. In the simplest one-factor case, the GIV is just *size-weighted minus equal-weighted outcomes*.
- **What determines power.** GIVs are strong when the economy is concentrated (a few large units matter) and when idiosyncratic shocks are volatile relative to aggregate shocks.
- **Why it matters.** The paper adds a general-purpose identification strategy to macroeconomics, finance, and IO settings where standard instruments are scarce but granular shocks are economically important.
- **Scope.** The paper is primarily a theory-and-methodology paper. It formalizes the estimator, derives optimal weighting and asymptotic theory, discusses robustness and weak-instrument issues, and then shows how the procedure extends to more general environments.

2 Data and measurement (key objects)

2.1 Economic objects rather than a single dataset

- **Unit of observation.** There are N entities indexed by i and T time periods indexed by t .
- **Observed variables.** The econometrician observes outcomes y_{it} , an aggregate price or endogenous variable p_t , a size vector $S = (S_i)_{i=1}^N$ with $\sum_i S_i = 1$, and possibly controls.
- **Unobserved variables.** Aggregate shocks h_t , equation error ε_t , and idiosyncratic shocks u_{it} .
- **Granularity.** The size weights S_i matter because a shock to a large entity affects aggregate outcomes, whereas shocks to tiny entities do not.
- **Equal-weighted vs. size-weighted aggregates.** The paper repeatedly compares

$$y_{Et} \equiv \frac{1}{N} \sum_{i=1}^N y_{it}, \quad y_{St} \equiv \sum_{i=1}^N S_i y_{it}.$$

These are the building blocks of the simplest GIV.

2.2 Factor structure and idiosyncratic shocks

- **Common factors.** The demand-side disturbance is modeled as a factor structure $l_i h_t$, where l_i is the loading of entity i on aggregate shocks.
- **Idiosyncratic shock.** The key orthogonality assumption is that u_{it} is uncorrelated with aggregate shocks and controls.
- **Practical meaning.** A valid idiosyncratic shock can be a productivity shock, a demand shock, or more subtly, an unexpected change in an entity's exposure to a common shock, as long as it is orthogonal to aggregate disturbances.

2.3 The object to be constructed: the granular IV

- **Generic form.** The GIV is a weighted sum of outcomes or residuals,

$$z_t = \Gamma' y_t,$$

where the weight vector Γ is chosen to load on size and to be orthogonal to common factors.

- **Interpretation.** Once common components are removed, z_t becomes a weighted sum of idiosyncratic shocks. This is the primitive exogenous variation used for identification.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline supply–demand system

$$p_t = w y_{St} + C_t^p m^p + \varepsilon_t, \tag{1}$$

$$y_{it} = f_d p_t + C_{it}^y m^y + l_i h_t + u_{it}. \tag{2}$$

- **Left-hand side.** p_t is the aggregate endogenous variable, interpreted in the motivating example as the log price of a good.
- **Aggregate demand object.** $y_{St} = \sum_i S_i y_{it}$ is the size-weighted aggregate outcome.
- **Key parameter w .** This is the sensitivity of price to aggregate demand. In the motivating supply–demand example, the supply elasticity is $f_s = 1/w$.
- **Key parameter f_d .** This is the demand elasticity, assumed homogeneous in the baseline model.
- **Identification problem.** OLS fails because y_{St} is correlated with ε_t in (1) and p_t is correlated with h_t in (2).

3.2 Exogeneity condition for idiosyncratic shocks

$$u_t \perp h_t, \varepsilon_t, C_t^y, C_t^p. \tag{3}$$

- **Interpretation.** The entire strategy hinges on isolating shocks that move large units but are orthogonal to aggregate shocks and controls.

- **Economic content.** This is what turns granularity into identification rather than mere descriptive concentration.

3.3 Construction of the GIV

Choose a vector $\Gamma \in \mathbb{R}^N$ such that

$$\Gamma' l = 0, \quad \Gamma' S \neq 0. \quad (4)$$

Then define

$$z_t \equiv \Gamma' y_t = \sum_{i=1}^N \Gamma_i y_{it}. \quad (5)$$

Using the demand equation,

$$z_t = \Gamma' u_t. \quad (6)$$

- **Why (4) matters.** Orthogonality to factor loadings strips out the common component $l_i h_t$.
- **Why (5)–(6) matter.** The instrument is observable because it is built from y_{it} , but econometrically it behaves like a weighted sum of unobserved idiosyncratic shocks.
- **Exogeneity.** From (3), z_t is orthogonal to aggregate shocks and the structural error.
- **Relevance.** Because $\Gamma' S \neq 0$, the instrument covaries with the size-weighted aggregate outcome.

3.4 Moment conditions for IV estimation

$$\mathbb{E}[(p_t - w y_{St}) z_t] = 0, \quad (9)$$

$$\mathbb{E}[u_{E \sim t} u_{\Gamma t}] = 0, \quad i' E \sim = 1, \quad (10)$$

$$\mathbb{E}[(y_{E \sim t} - f_d p_t) z_t] = 0. \quad (11)$$

$$\mathbb{E}[p_t z_t] \neq 0. \quad (12)$$

- **What is estimated.** The first moment identifies w ; the second set of moments identifies f_d .
- **Quasi-equal weights.** $E \sim$ is chosen so that the equal-weighted demand residual is orthogonal to the GIV-induced residual. Under homoskedasticity, this reduces to simple equal weighting.
- **Key intuition.** An idiosyncratic shock to a large unit changes price, and the induced price response identifies how other units react.

3.5 Pass-through objects

$$M \equiv \frac{1}{1 - w f_d}, \quad m \equiv w M. \quad (13)$$

- M . Pass-through from idiosyncratic shocks to the aggregate quantity.

- *m*. Pass-through from idiosyncratic shocks to price.
- **Why useful.** These are often the economically interesting reduced-form objects, even when the elasticities themselves are not the final target.

3.6 Optimal GIV weights

$$Q \equiv I - l(l'l)^{-1}l', \quad (16)$$

$$\Gamma^{*l} = S'Q. \quad (17)$$

- **Interpretation.** The optimal GIV loads on size as much as possible while being orthogonal to common factors.
- **Economic intuition.** Each entity's contribution to aggregate prices is proportional to size, but the instrument must discard the component aligned with aggregate shocks.

3.7 Recoverable factors

Start from the factor model

$$Y_t = lh_t + u_t, \quad h_t \perp u_t, \quad (19)$$

with

$$R_l \equiv (l'l)^{-1}l', \quad Q_l \equiv I - lR_l, \quad (20)$$

and define

$$\tilde{h}_t \equiv h_t + R_l u_t, \quad \tilde{u}_t \equiv Q_l u_t. \quad (21)$$

Then

$$Y_t = l\tilde{h}_t + \tilde{u}_t, \quad \tilde{h}_t, \tilde{u}_t \perp, \quad (22)$$

and the recoverable factors are

$$\tilde{h}_t = R_l Y_t, \quad \tilde{u}_t = Q_l Y_t. \quad (23)$$

- **Idea.** Although the original factor h_t cannot be recovered without error when N is fixed, a rotated object $\tilde{h}_t$ can be.
- **Why important.** This gives an operational procedure to purge common components and estimate the GIV using actual data.

3.8 Estimation with controls

The key observable transformations are

$$z_t(m^y) \equiv \Gamma'(y_t - C_t^y m^y), \quad (27)$$

and moment conditions

$$\sum_t (-\tilde{y}_t + \tilde{l}\tilde{h}_t(m^y) + \tilde{C}_t^y m^y)' \tilde{C}_t^y = 0, \quad (28)$$

$$\sum_t (-p_t + wy_{St} + b^p \tilde{h}_t(m^y) + C_t^p m^p)(z_t(m^y), \tilde{h}_t(m^y)', C_t^{p'}) = 0, \quad (29)$$

$$\sum_t (-y_{E \sim t} + f_d p_t + b^y \tilde{h}_t(m^y) + C_{E \sim t}^y m^y)(z_t(m^y), \tilde{h}_t(m^y)', C_{E \sim t}^{y'}) = 0. \quad (30)$$

- **Purpose.** These moments show how to estimate the GIV and the elasticities after partialling out both observed controls and latent factors.
- **Takeaway.** The procedure is implementable even when the econometrician must first recover factors and residuals from cross-sectional structure.

3.9 OLS use of the GIV

The paper also shows that some reduced-form objects can be estimated by OLS once the GIV is constructed:

$$p_t = mz_t + b^p c_t + e_t^p, \quad (36)$$

$$y_{St} = Mz_t + b^y c_t + e_t^y. \quad (37)$$

- **Interpretation.** If the researcher wants pass-through coefficients m and M , no IV ratio is needed; OLS of price and aggregate demand on the GIV is enough.
- **Why useful.** This is especially attractive because weak-IV concerns are less severe for these OLS estimands: low first-stage power simply shows up as large standard errors.

3.10 Simple one-factor case

In the simplest case,

$$p_t = wy_{St} + \varepsilon_t, \quad (38)$$

$$y_{it} = f_d p_t + h_t + u_{it}. \quad (39)$$

Then the optimal GIV collapses to

$$z_t = y_{St} - y_{Et} = u_{St} - u_{Et}. \quad (43)$$

The asymptotic precision depends on the excess Herfindahl

$$h \equiv \left(\sum_{i=1}^N S_i^2 - \frac{1}{N} \right)^{1/2}. \quad (45)$$

- **Most intuitive takeaway in the paper.** In a one-factor world, the instrument is literally *size-weighted minus equal-weighted outcomes*.
- **Power formula.** Precision rises with concentration h and with the volatility of idiosyncratic shocks.

3.11 Heterogeneous demand elasticities

$$f_{di} = x_i \tilde{f}_d = \sum_{\ell=1}^k x_{i\ell} \tilde{f}_{d\ell}. \quad (47)$$

$$\mathbb{E}[(\tilde{y}_{\ell t} - \tilde{f}_{d\ell} p_t) z_t] = 0, \quad \ell = 1, \dots, k. \quad (48)$$

- **Interpretation.** Elasticities can vary systematically with observable characteristics x_i .
- **Importance.** The paper is not limited to a homogeneous-coefficient world; it provides a practical extension that recovers a vector of elasticity parameters.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **Comparison.** Compare changes in outcomes caused by large idiosyncratic shocks to big entities, after removing their common factor exposure.
- **Source of exogenous variation.** The instrument is a weighted sum of idiosyncratic shocks, not an aggregate policy shock imported from outside the system.
- **Exclusion restriction.** Once common factors are removed, the GIV affects the endogenous regressor only through the targeted granular shocks.
- **Relevance.** Large units matter for aggregates. If a few units are big enough, their shocks move prices and quantities enough to identify elasticities.

4.2 What threatens identification

- **Main threat.** Failure to separate common from idiosyncratic shocks. If the econometrician leaves aggregate shocks inside the estimated residuals, the GIV will be contaminated and biased.
- **Other concerns.** Weak concentration, low idiosyncratic volatility, instability of factor loadings, heteroskedasticity, and heterogeneous elasticities across entities.

4.3 Diagnostics and robustness checks proposed in the paper

- **Add estimated factors.** Enrich the factor structure and test whether estimates remain stable.
- **Narrative check.** Inspect the largest idiosyncratic shocks in historical context to verify that they really are idiosyncratic.
- **Remove sporadic factor realizations.** If a large "idiosyncratic" shock coincides with an unusual aggregate event, drop that date.
- **Overidentification test.** Split large units into groups (for example, developed and developing countries), construct separate GIVs, and compare the resulting estimates.

- **Subset robustness.** The IV estimators remain valid even if the researcher constructs the GIV using only a subset of the large entities.
- **Mismeasured size distribution.** IV elasticities remain consistent under size-weight misspecification, though OLS pass-through estimates may then be biased.

4.4 Weak-instrument logic

- **Precision condition.** A strong GIV needs both a concentrated economy and large idiosyncratic shocks relative to aggregate shocks.
- **Operational lesson.** Researchers can run an ex ante power analysis using concentration and the volatility of idiosyncratic shocks before collecting large new datasets.
- **2SLS advice.** The paper discusses conventional first-stage diagnostics and emphasizes that weak-instrument concerns apply especially to ratio-based IV estimands such as w and f_d .

4.5 Relation to Bartik instruments

- **Complement, not substitute.** Bartik is useful when there is cross-sectional exposure to common shocks; GIV is useful when a few large idiosyncratic shocks themselves move aggregates.
- **Takeaway.** The right instrument depends on the economic environment. GIV is tailored to granular economies.

5 Tables and figures

5.1 No main empirical tables or figures

Read:

- This is mainly a theory and methodology paper. The main "exhibits" are the sequence of equations, propositions, and corollaries rather than regression tables or plots.
- The paper's logic is best read in the following order: baseline model $\rightarrow$ optimal GIV $\rightarrow$ controls/recoverable factors $\rightarrow$ one-factor intuition $\rightarrow$ robustness/extensions.

5.2 Propositions 1–3: Baseline estimator and optimal weighting

Read:

- Proposition 1 establishes consistency of the IV estimators for w and f_d .
- Proposition 2 gives asymptotic variance formulas and shows how precision depends on the covariance between the GIV and aggregate idiosyncratic demand.
- Proposition 3 is the core design result: the optimal GIV weights are the size vector projected orthogonally to common-factor loadings.

5.3 Proposition 4 and Lemma 1: Controls and recoverable factors

Read:

- Lemma 1 is the technical device that makes the method feasible with fixed N : although the raw factor is not recoverable, a transformed factor and residual are.
- Proposition 4 then converts the idea into a workable estimator with observed controls and latent-factor controls.

5.4 Corollaries 1–2: Intuition in the single-factor case

Read:

- Corollary 1 gives the cleanest economic intuition: $z_t = y_{St} - y_{Et}$.
- Corollary 2 links estimator precision to the excess Herfindahl and the variance of idiosyncratic shocks. This is the most important result for thinking about whether GIVs will be useful in a given application.

5.5 Proposition 5 and Section IV: OLS use and generalizations

Read:

- Proposition 5 shows that pass-through objects can be estimated by OLS once the GIV is formed.
- Section IV demonstrates that the method extends naturally to heterogeneous elasticities, estimated factor loadings, heteroskedasticity, time-varying factor exposures, multidimensional outcomes, and even VAR identification.

6 Short summary

- **Conclusion.** In granular economies, idiosyncratic shocks to large units can be isolated and aggregated into valid instruments for endogenous aggregate variables.
- **Intuition.** The instrument is built from the part of large units' outcomes that is idiosyncratic rather than common, so it is both relevant and exogenous.
- **Empirical lesson.** Before using GIVs, researchers should check two things: whether the economy is concentrated enough, and whether they can credibly purge common factors.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper provides a new, general methodology for constructing instruments from granular idiosyncratic shocks.
- **Theoretical contribution.** It derives consistency, optimal weighting, and asymptotic variance formulas for both IV and OLS estimators built from these shocks.

- **Economic contribution.** It clarifies that identification can come from large units inside the economy, not only from external policy or natural-experiment shocks.
- **Credibility conditions.** The procedure is credible when the researcher can separate common and idiosyncratic components and when concentration plus idiosyncratic volatility generate sufficient first-stage strength.
- **Generality.** The method extends beyond the simple supply–demand model to richer factor structures, heterogeneous elasticities, heteroskedastic shocks, multidimensional outcomes, and external-instrument identification for dynamic systems.

7.2 Economic intuition

- A shock to a tiny unit is irrelevant for aggregates, but a shock to a very large unit is not. That is the basic source of relevance.
- Aggregate shocks contaminate any naive use of large-unit outcomes. That is why the instrument must be orthogonalized with respect to factor loadings.
- The method works best when there are a few dominant players and the granular shocks are noisy in an economically useful way, so that they move aggregates but remain plausibly exogenous.

7.3 Takeaway

This paper is best understood as a clean methodology paper for settings in which aggregate fluctuations have granular origins. The central practical message is simple: if a few large units matter enough, and if one can isolate their idiosyncratic shocks from aggregate forces, then those shocks can become powerful instruments. For empirical researchers, the value of the paper is not only the GIV formula itself, but also the checklist it provides: identify the large players, purge common factors carefully, test power, and verify the largest shocks narratively.